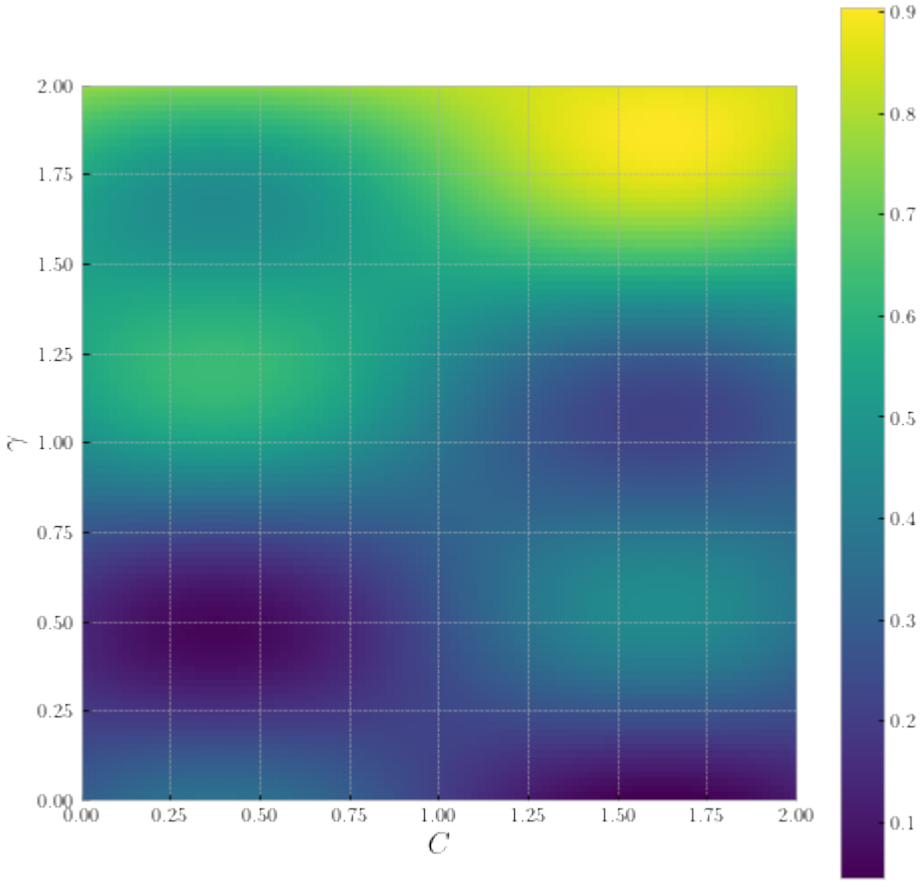


Figure 3.17 Accuracy of an SVM model on a test data set, as a function of the penalty parameter c and the RBF kernel parameter . The function changes more quickly with respect to than to c.

1. Randomly draw 100 data points from the domain $[0, 2]^2$. This will be used as our training data.
2. Implement a GP model with a constant mean function and a Matérn 2.5 kernel with an output scale implemented as a `gpytorch.kernels.ScaleKernel` object.
3. Don't specify the `ard_num_dims` parameter when initializing the kernel object, or set the parameter to `None`. This will create a GP model without ARD.
4. Train the hyperparameters of the GP model using gradient descent, and inspect the length scale after training.
5. Redefine the GP model class, this time setting `ard_num_dims = 2`. Retraining the GP model with gradient descent, and verify that the two length scales have significantly different values.

The solution is included in `CH03/03 - Exercise.ipynb`.

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